

Sequences and Series

Date_____ Period____

Find the next three terms in each sequence.

1) 1, 3, 6, 10, 15, ...

2) -4, -8, -16, -32, -64, ...

Find the first four terms in each sequence.

3) $a_n = 3 \cdot 4^{n-1}$

4) $a_n = -\frac{6}{n+3}$

5) $a_n = a_{n-1} \cdot -5$
 $a_1 = 0.4$

6) $a_n = a_{n-1} \cdot 3$
 $a_1 = 3$

Find the tenth term in each sequence.

7) $a_n = -\frac{15}{n}$

8) $a_n = a_{n-1} + n$
 $a_1 = -8$

Determine if each sequence converges or diverges.

9) -30, -14, -6, -2, 0, ...

10) 8, 108, 208, 308, 408, ...

11) $a_n = \frac{6}{n+3}$

12) $a_n = -4 \cdot (-4)^{n-1}$

13) $a_n = a_{n-1} \cdot 2$
 $a_1 = 3$

14) $a_n = \frac{2 + a_{n-1}}{2}$
 $a_1 = -30$